(Sn), germanium (Ge), and manganese (Mn)—these are called the *parent compounds*—at different ratios. The features are contained in the first four columns, which are the percentages of the parent compounds. The prediction target, mixing temperature, is in the last column, denoting the lowest temperature at which an alloy can form. The task is to predict mixing temperature given the compositional percentages of an alloy.

Table 2.2 Data from a materials discovery task. The features are the structure of a material expressed in percentages of parent compounds, and the prediction target is the mixing temperature.

% of Pb	% of Sn	% of Ge	% of Mn	Mixing temp.
0.50	0.50	0.00	0.00	192.08
0.33	0.33	0.33	0.00	258.30
0.00	0.50	0.50	0.00	187.24
0.00	0.33	0.33	0.33	188.54

There are multiple steps:

1. Create the four-dimensional data set included in Table 2.2.
2. Normalize the fifth column by subtracting the mean from all values and dividing the results by their variance.
3. Treat the first four columns as features and the fifth as labels. Train a Gaussian process on this data. You can reuse the GP model class we implemented in the chapter.
4. Create a test data set containing compositions having zero percentage of germanium and manganese. In other words, the test set is a grid over the unit square whose axes are percentages of lead and tin.

   The test set should look like the following PyTorch tensor.

```
tensor([[0.0000, 0.0000, 0.0000, 0.0000],
        [0.0000, 0.0100, 0.0000, 0.0000],
        [0.0000, 0.0200, 0.0000, 0.0000],
        ...,
        [1.0000, 0.9800, 0.0000, 0.0000],
        [1.0000, 0.9900, 0.0000, 0.0000],
        [1.0000, 1.0000, 0.0000, 0.0000]])
```

   Notice that the third and forth columns are all zeros.
5. Predict the mixing temperature on this test set. That is, compute the posterior mean and standard deviation of the normalized mixing temperature for every point in the test set.
6. Visualize the predictions. This involves showing the mean and standard deviation as heat maps in the same way as in Figure 2.19.

The solution is included in `CH02/02 - Exercise.ipynb`.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>